Chapter 12

Sequential Quadratic Programming

12.1 Lagrange-Newton Method

Consider the equality constrained optimization problem

$$\min_{x \in \Re^n} \quad f(x) \tag{12.1.1}$$

$$\text{s.t.} \quad c(x) = 0, \tag{12.1.2}$$

where $c(x) = (c_1(x), \cdots, c_m(x))^T \in \Re^n$. The Lagrangian function is

$$L(x, \lambda) = f(x) - \lambda^T c(x). \tag{12.1.3}$$

A point x is a KKT point of (12.1.1)-(12.1.2) if and only if there exists $\lambda \in \Re^m$ such that

$$\nabla_x L(x, \lambda) = \nabla f(x) - \nabla c(x)^T \lambda = 0, \tag{12.1.4}$$

$$\nabla_\lambda L(x, \lambda) = -c(x) = 0. \tag{12.1.5}$$

The nonlinear system (12.1.4)-(12.1.5) requires x to be a stationary point of the Lagrangian function. Therefore any method based on solving (12.1.4)-(12.1.5) can be called a Lagrange method. For a given iterate point $x_k \in \Re^n$ and an approximate Lagrange multiplier $\lambda_k \in \Re^m$, the Newton-Raphson step for solving (12.1.4)-(12.1.5) is $((\delta x)_k, (\delta\lambda)_k)$, which satisfies

$$\begin{pmatrix} W(x_k, \lambda_k) & -A(x_k) \\ -A(x_k)^T & 0 \end{pmatrix} \begin{pmatrix} (\delta x)_k \\ (\delta\lambda)_k \end{pmatrix} = - \begin{pmatrix} \nabla f(x_k) - A(x_k)\lambda_k \\ -c(x_k) \end{pmatrix}, \tag{12.1.6}$$

where

$$A(x_k) = \nabla c(x_k)^T, \tag{12.1.7}$$

$$W(x_k, \lambda_k) = \nabla^2 f(x_k) - \sum_{i=1}^{m} (\lambda_k)_i \nabla^2 c_i(x_k). \tag{12.1.8}$$

Consider the penalty function

$$P(x, \lambda) = \|\nabla f(x) - A(x)\lambda\|_2^2 + \|c(x)\|_2^2; \tag{12.1.9}$$

it is easy to show that $(\delta x)_k$ and $(\delta\lambda)_k$ defined by (12.1.6) satisfy that

$$((\delta x)_k^T, (\delta\lambda)_k^T)\nabla P(x_k, \lambda_k) = -2P(x_k, \lambda_k) \le 0. \tag{12.1.10}$$

Here ∇P is the gradient of P in the space (x, λ). The method given below is based on (12.1.6), hence it is called the Lagrange-Newton method.

Algorithm 12.1.1 *(Lagrange-Newton Method)*

Step 1. *Given* $x_1 \in \Re^n$, $\lambda_1 \in \Re^m$, $\beta \in (0, 1)$, $\epsilon \ge 0$, $k := 1$;

Step 2. *Compute* $P(x_k, \lambda_k)$; *if* $P(x_k, \lambda_k) \le \epsilon$ *then stop;*
solving (12.1.6) obtaining $(\delta x)_k$ *and* $(\delta\lambda)_k$;
$\alpha = 1$;

Step 3. *if*

$$P(x_k + \alpha(\delta x)_k, \lambda_k + \alpha(\delta\lambda)_k) \le (1 - \beta\alpha)P(x_k, \lambda_k), \tag{12.1.11}$$

then go to Step 4;
$\alpha = \alpha/4$, *go to Step 3;*

Step 4. $x_{k+1} = x_k + \alpha(\delta x)_k$; $\lambda_{k+1} = \lambda_k + \alpha(\delta\lambda)_k$;
$k := k + 1$; *go to Step 2.* $\square$

For the above algorithm, we have the following convergence result.

Theorem 12.1.2 *Assume that* $f(x)$ *and* $c(x)$ *are twice continuously differentiable. If the matrix*

$$\begin{bmatrix} W(x_k, \lambda_k) & -A(x_k) \\ -A(x_k)^T & 0 \end{bmatrix}^{-1} \tag{12.1.12}$$

is uniformly bounded, then any accumulation point of $\{(x_k, \lambda_k)\}$ *generated by Algorithm 12.1.1 is a root of* $P(x, \lambda) = 0$.

Proof. Suppose that the theorem is not true, i.e., suppose $(\bar{x}, \bar{\lambda})$ is an accumulation point of $\{(x_k, \lambda_k)\}$ and

$$P(\bar{x}, \bar{\lambda}) > 0. \tag{12.1.13}$$

Then there exists a subset $K_0 \subseteq \{1, 2, \cdots\}$ which has infinitely many elements and satisfies that

$$\lim_{\substack{k\in K_0\\k\to\infty}} x_k = \bar{x}, \qquad \lim_{\substack{k\in K_0\\k\to\infty}} \lambda_k = \bar{\lambda}. \tag{12.1.14}$$

From the line search condition (12.1.11), we can see that

$$P(x_{k+1}, \lambda_{k+1}) \le (1 - \beta\alpha_k) P(x_k, \lambda_k). \tag{12.1.15}$$

It follows from (12.1.13)-(12.1.15) that

$$\lim_{\substack{k\in K_0\\k\to\infty}} \alpha_k = 0. \tag{12.1.16}$$

Therefore we have that

$$P(x_k + \hat{\alpha}_k(\delta x)_k, \lambda_k + \hat{\alpha}_k(\delta\lambda)_k) > (1 - \beta\hat{\alpha}_k) P(x_k, \lambda_k) \tag{12.1.17}$$

for all sufficiently large $k \in K_0$, where $\hat{\alpha}_k = 4\alpha_k \in (0, 1)$. Let $(\overline{\delta x}, \overline{\delta\lambda})$ be the solution of

$$\begin{pmatrix} W(\bar{x}, \bar{\lambda}) & -A(\bar{x}) \\ -A(\bar{x})^T & 0 \end{pmatrix} \begin{pmatrix} \delta x \\ \delta\lambda \end{pmatrix} = - \begin{pmatrix} \nabla f(\bar{x}) - \nabla c(\bar{x})^T \bar{\lambda} \\ c(\bar{x}) \end{pmatrix}. \tag{12.1.18}$$

Because $\hat{\alpha}_k \to 0$, we can show that

$$\lim_{\substack{k\in K_0\\k\to\infty}} \frac{P(\bar{x} + \hat{\alpha}_k\overline{\delta x}, \bar{\lambda} + \hat{\alpha}_k\overline{\delta\lambda}) - P(\bar{x}, \bar{\lambda})}{\hat{\alpha}_k} = -2P(\bar{x}, \bar{\lambda}) < -P(\bar{x}, \bar{\lambda}). \tag{12.1.19}$$

From the uniform boundedness of (12.1.12) and the fact that $(x_k, \lambda_k) \to (\bar{x}, \bar{\lambda})(k \in K_0)$, it follows that $((\delta x)_k, (\delta\lambda)_k) \to (\overline{\delta x}, \overline{\delta\lambda})$. Therefore, for sufficiently large $k \in K_0$ we have that

$$\frac{P(x_k + \hat{\alpha}_k(\delta x)_k, \lambda_k + \hat{\alpha}_k(\delta\lambda)_k) - P(x_k, \lambda_k)}{\hat{\alpha}_k} \le -P(x_k, \lambda_k). \tag{12.1.20}$$

Because $\beta < 1$, (12.1.20) contradicts (12.1.17). This implies that the theorem is true. $\square$

Theorem 12.1.3 *Assume that $f(x)$ and $c(x)$ are twice continuously differentiable. If the matrix (12.1.12) is uniformly bounded, then any accumulation point of $\{x_k\}$ generated by Algorithm 12.1.1 is a KKT point of (12.1.1)-(12.1.2).*

Proof. If the theorem is not true, it follows from the monotonicity of $P(x_k, \lambda_k)$ that

$$\lim_{k\to\infty} P(x_k, \lambda_k) > 0. \tag{12.1.21}$$

This limit and condition (12.1.11) imply that

$$\prod_{k=1}^{\infty}(1-\beta\alpha_k) > 0. \tag{12.1.22}$$

The above relation indicates that

$$\sum_{k=1}^{\infty}\alpha_k < +\infty. \tag{12.1.23}$$

Because

$$\begin{bmatrix} W(x_k,\lambda_k) & -A(x_k) \\ -A(x_k)^T & 0 \end{bmatrix} \begin{bmatrix} (\delta x)_k \\ \lambda_k + (\delta\lambda)_k \end{bmatrix} = \begin{bmatrix} -\nabla f(x_k) \\ c(x_k) \end{bmatrix}, \tag{12.1.24}$$

there exists a positive constant $\gamma > 0$ such that

$$\|(\delta x)_k\| + \|\lambda_k + (\delta\lambda)_k\| \le \gamma(\|\nabla f(x_k)\| + \|c(x_k)\|). \tag{12.1.25}$$

Let $\bar{x}$ be any accumulation point of $\{x_k\}$. Define the set

$$S_\delta = \{x \,|\, \|x-\bar{x}\| \le \delta\}, \tag{12.1.26}$$

where $\delta > 0$ is any given positive constant. From (12.1.25) we know that there exists a constant $\eta > 0$ such that

$$\|(\delta x)_k\| \le \eta \tag{12.1.27}$$

for all $x_k \in S_\delta$. It follows from (12.1.23) that there exists $\bar{k}$ such that

$$\sum_{k=\bar{k}}^{\infty}\alpha_k < \frac{\delta}{2\eta}. \tag{12.1.28}$$

Because $\bar{x}$ is an accumulation point of $\{x_k\}$, there exists $\hat{k} > \bar{k}$ such that

$$\|x_{\hat{k}} - \bar{x}\| < \frac{\delta}{2}. \tag{12.1.29}$$

From (12.1.27)-(12.1.29) and the fact that $\|x_{k+1} - x_k\| = \alpha_k \|(\delta x)_k\|$ we have that

$$x_k \in S_\delta, \quad \forall k \geq \hat{k}. \tag{12.1.30}$$

Therefore (12.1.27) holds for all $k \geq \hat{k}$. Thus, it follows from (12.1.23) that

$$\lim_{k\to\infty} x_k = \bar{x}. \tag{12.1.31}$$

This relation and the last theorem imply that there are no accumulation points of $\{(x_k, \lambda_k)\}$, which shows that

$$\lim_{k\to\infty} \|\lambda_k\| = \infty. \tag{12.1.32}$$

Hence, it follows from (12.1.32) and (12.1.25) that

$$\begin{aligned} \|\lambda_{k+1}\| &= \|\lambda_k + \alpha_k(\delta\lambda)_k\| \\ &= \|(1-\alpha_k)\lambda_k + \alpha_k(\lambda_k + (\delta\lambda)_k)\| \\ &= (1-\alpha_k)\|\lambda_k\| + O(\alpha_k) < \|\lambda_k\| \end{aligned} \tag{12.1.33}$$

holds for all sufficiently large k, which contradicts (12.1.32). This completes our proof. $\square$

About the convergence rate of Algorithm 12.1.1, we have the following result.

Theorem 12.1.4 *Assume that the sequence $\{x_k\}$ generated by Algorithm 12.1.1 converges to x^*, if $f(x)$ and $c(x)$ are three times continuously differentiable near x^*, $A(x^*)$ is full column rank, and the second-order sufficient condition is satisfied at x^*, then $\lambda_k \to \lambda^*$, and*

$$\left\| \begin{pmatrix} x_{k+1} - x^* \\ \lambda_{k+1} - \lambda^* \end{pmatrix} \right\| = O\left(\left\| \begin{pmatrix} x_k - x^* \\ \lambda_k - \lambda^* \end{pmatrix} \right\|^2 \right). \tag{12.1.34}$$

Proof. Because Algorithm 12.1.1 is the Newton-Raphson method for (12.1.4)-(12.1.5), and because the second-order sufficient condition implies that the matrix

$$\begin{bmatrix} W(x^*, \lambda^*) & -A(x^*) \\ -A(x^*)^T & 0 \end{bmatrix} \tag{12.1.35}$$

is nonsingular, we have that

$$\left\| \begin{pmatrix} x_k + (\delta x)_k - x^* \\ \lambda_k + (\delta\lambda)_k - \lambda^* \end{pmatrix} \right\| = O\left(\left\| \begin{pmatrix} x_k - x^* \\ \lambda_k - \lambda^* \end{pmatrix} \right\|^2 \right) \tag{12.1.36}$$

for all sufficiently large k. The above relation, and the fact that $f(x)$ and $c(x)$ are three times continuously differentiable imply that (12.1.11) holds for $\alpha = 1$. Therefore (12.1.34) holds. $\square$

It should be pointed out that (12.1.34) is not equivalent to the usual quadratic convergence, which is

$$\|x_{k+1} - x^*\| = O(\|x_k - x^*\|^2). \tag{12.1.37}$$

For the analysis of the convergence rate of the iterates $\{x_k\}$, we need the following result.

Lemma 12.1.5 *Under the assumptions of Theorem 12.1.4, we have that*

$$\epsilon_{k+1} = O(\|x_k - x^*\|\epsilon_k), \tag{12.1.38}$$

where

$$\epsilon_k = \|x_k - x^*\| + \|\lambda_k - \lambda^*\|. \tag{12.1.39}$$

Proof. From the proof of Theorem 12.1.4 we see that $\alpha_k = 1$ for all sufficiently large k. Therefore it follows from the definitions of $(\delta x)_k$ and $(\delta\lambda)_k$ that

$$\begin{aligned} \begin{bmatrix} W(x_k, \lambda_k) & -A(x_k) \\ -A(x_k)^T & 0 \end{bmatrix} \begin{bmatrix} x_{k+1} - x^* \\ \lambda_{k+1} - \lambda^* \end{bmatrix} &= \begin{bmatrix} -\nabla f(x_k) + A(x_k)\lambda_k \\ c(x_k) \end{bmatrix} \\ &\quad + \begin{bmatrix} W(x_k, \lambda_k)(x_k - x^*) - A(x_k)(\lambda_k - \lambda^*) \\ -A(x_k)^T(x_k - x^*) \end{bmatrix} \\ &= \begin{bmatrix} (A(x^*) - A(x_k))(\lambda_k - \lambda^*) + O(\|x_k - x^*\|^2) \\ O(\|x_k - x^*\|^2) \end{bmatrix} \\ &= \begin{bmatrix} O(\|x_k - x^*\|[\|x_k - x^*\| + \|\lambda_k - \lambda^*\|]) \\ O(\|x_k - x^*\|^2) \end{bmatrix} \\ &= O(\|x_k - x^*\|\epsilon_k). \end{aligned} \tag{12.1.40}$$

The above relation and the nonsingularity of matrix (12.1.35) show that the lemma holds. $\square$

Theorem 12.1.6 *Under the assumptions of Theorem 12.1.4, the sequence* $\{x_k\}$ *converges to* x^* *superlinearly and*

$$\|x_{k+1} - x^*\| = o\left(\|x_k - x^*\| \prod_{j=1}^{p} \|x_{k-j} - x^*\|\right) \tag{12.1.41}$$

holds for any given positive integer p.

Proof. It follows from (12.1.38) that $\{x_k\}$ converges to x^* superlinearly. For any given positive integer p, applying (12.1.38) recursively we obtain that

$$\begin{aligned}\|x_{k+1} - x^*\| &= O(\epsilon_{k+1}) = O(\|x_k - x^*\|\epsilon_k) \\ &= O(\|x_k - x^*\| \, \|x_{k-1} - x^*\|\epsilon_{k-1}) \\ &= O\left(\|x_k - x^*\| \prod_{j=1}^{p} \|x_{k-j} - x^*\|\epsilon_{k-p}\right) \\ &= o\left(\|x_k - x^*\| \prod_{j=1}^{p} \|x_{k-j} - x^*\|\right). \end{aligned} \tag{12.1.42}$$

Therefore the theorem is true. $\Box$

One of the most important contributions of the Lagrange-Newton method is the development of the sequential quadratic programming method based on it. Sequential quadratic programming algorithms are the most important algorithms for solving medium and small scale nonlinear constrained optimization problems.

Setting $\bar{\lambda}_k = \lambda_k + (\delta\lambda)_k$, we can write (12.1.6) in the following equivalent form

$$W(x_k, \lambda_k)(\delta x)_k + \nabla f(x_k) = A(x_k)[\lambda_k + (\delta\lambda)_k], \tag{12.1.43}$$

$$c(x_k) + A(x_k)^T(\delta x)_k = 0, \tag{12.1.44}$$

which is just, in matrix form,

$$\begin{bmatrix} W(x_k, \lambda_k) & -A(x_k) \\ -A(x_k)^T & 0 \end{bmatrix} \begin{bmatrix} (\delta x) \\ \bar{\lambda} \end{bmatrix} = \begin{bmatrix} -g(x_k) \\ c(x_k) \end{bmatrix} \tag{12.1.45}$$

with solution $(\delta x)_k$ and $\bar{\lambda}_k$. Then x_{k+1} is given by

$$x_{k+1} = x_k + (\delta x)_k. \tag{12.1.46}$$

It is easy to show that the above system (12.1.45) is a KKT condition of the quadratic programming subproblem

$$\min_{d\in\Re^n} \quad d^T\nabla f(x_k) + \frac{1}{2}d^T W(x_k,\lambda_k)d, \tag{12.1.47}$$

$$\text{s.t.} \quad c(x_k) + A(x_k)^T d = 0 \tag{12.1.48}$$

with $((\delta x)_k, \bar{\lambda}_k)$ being the corresponding KKT pair thereof. Therefore, the Lagrange-Newton method can be viewed as a method that solves the quadratic programming subproblem (12.1.47)-(12.1.48) successively.

12.2 Wilson-Han-Powell Method

In this section we present a sequential quadratic programming method, which was proposed by Han [169]. The method is based on the Lagrange-Newton method discussed in the previous section. In each iteration the matrix $W(x_k,\lambda_k)$ is replaced by a matrix B_k. Because the Lagrange-Newton method was first considered by Wilson [349], and because Han's method was modified and analyzed by Powell [268], the method presented in this section is often called the Wilson-Han-Powell method.

Consider nonlinearly constrained optimization problem (8.1.1)-(8.1.3), Similar to (12.1.47)-(12.1.48), we construct the following subproblem

$$\min_{d\in\Re^n} \quad g_k^T d + \frac{1}{2}d^T B_k d, \tag{12.2.1}$$

$$\text{s.t.} \quad a_i(x_k)^T d + c_i(x_k) = 0, \; i \in E, \tag{12.2.2}$$

$$a_i(x_k)^T d + c_i(x_k) \geq 0, \quad i \in I, \tag{12.2.3}$$

where

$$A(x_k) = [a_1(x_k), \cdots, a_m(x_k)] = \nabla c(x_k)^T, \tag{12.2.4}$$

$g_k = g(x_k) = \nabla f(x_k)$, $E = \{1, 2, \cdots m_e\}$, $I = \{m_e+1, \cdots, m\}$, and $B_k \in \Re^{n\times n}$ is an approximation to the Hessian matrix of the Lagrangian function. Let d_k be a solution of (12.2.1)-(12.2.3). The vector d_k is the search direction in the k-th iteration by the Wilson-Han-Powell method. Let λ_k be the corresponding Lagrange multiplier of (12.2.1)-(12.2.3) (just like $\bar{\lambda}_k$ in the previous section), then it follows that

$$g_k + B_k d_k = A(x_k)\lambda_k, \tag{12.2.5}$$

$$(\lambda_k)_i \geq 0, \quad i \in I, \tag{12.2.6}$$

$$(\lambda_k)_i[c_i(x_k) + a_i(x_k)^T d_k] = 0, \quad i \in I. \tag{12.2.7}$$

A very good property of d_k is that it is a descent direction of many penalty functions. For example, considering the L_1 exact penalty function, we have the following result.

Lemma 12.2.1 *Let d_k be a KKT point of (12.2.1)-(12.2.3) and λ_k be the corresponding Lagrange multiplier. Consider the L_1 penalty function*

$$P(x, \sigma) = f(x) + \sigma\|c^{(-)}(x)\|_1, \tag{12.2.8}$$

where $c^{(-)}(x)$ is defined by (10.1.2)-(10.1.3). Then we have that

$$P'_\alpha(x_k + \alpha d_k, \sigma)|_{\alpha=0} \le -d_k^T B_k d_k - \sigma\|c^{(-)}(x_k)\|_1 + \lambda_k^T c(x_k). \tag{12.2.9}$$

If $d_k^T B_k d_k > 0$ and $\sigma \ge \|\lambda_k\|_\infty$, then d_k is a descent direction of the penalty function (12.2.8) at x_k.

Proof. By Taylor expression and using convexity of $\|(c + Ad)^{(-)}\|_1$, we have that

$$\begin{aligned}
P'_\alpha(x_k + \alpha d_k, \sigma)|_{\alpha=0} &= \lim_{\alpha\to 0_+} \frac{P(x_k + \alpha d_k) - P(x_k)}{\alpha} \\
&= g_k^T d_k + \lim_{\alpha\to 0_+} \sigma \frac{\|[c(x_k) + \alpha A(x_k)^T d_k]^{(-)}\|_1 - \|c^{(-)}(x_k)\|_1}{\alpha} \\
&\le g_k^T d_k + \sigma[\|(c(x_k) + A(x_k)^T d_k)^{(-)}\|_1 - \|c^{(-)}(x_k)\|_1] \\
&= g_k^T d_k - \sigma\|c^{(-)}(x_k)\|_1.
\end{aligned} \tag{12.2.10}$$

It follows from (12.2.5) and (12.2.7) that

$$g_k^T d_k = -d_k^T B_k d_k + \lambda_k^T c(x_k). \tag{12.2.11}$$

Therefore (12.2.9) follows from (12.2.10) and (12.2.11).

Because λ_k satisfies (12.2.6), it follows from the definition of $c^{(-)}(x)$ that

$$\lambda_k^T c(x_k) \le \sum_{i=1}^m |(\lambda_k)_i| \; |c_i^{(-)}(x_k)|. \tag{12.2.12}$$

Substituting the above inequality into (12.2.9), and using the assumptions that $d_k^T B_k d_k > 0$ and $\sigma \ge \|\lambda_k\|_\infty$, we have that

$$P'_\alpha(x_k + \alpha d_k, \sigma)|_{\alpha=0} \le -d_k^T B_k d_k - \sum_{i=1}^m (\sigma - |(\lambda_k)_i|)|c_i^{(-)}(x_k)| < 0. \tag{12.2.13}$$

This shows that the lemma is true. □

The following algorithm is the sequential quadratic programming algorithm proposed by Han [169].

Algorithm 12.2.2

Step 1. *Given* $x_1 \in \Re^n$, $\sigma > 0$, $\delta > 0$, $B_1 \in \Re^{n\times n}$, $\epsilon \geq 0$, $k := 1$;

Step 2. *Solve (12.2.1)-(12.2.3) giving* d_k;
if $\|d_k\| \leq \epsilon$ *then stop;*
find $\alpha_k \in [0, \delta]$ *such that*

$$P(x_k + \alpha_k d_k, \sigma) \leq \min_{0\leq\alpha\leq\delta} P(x_k + \alpha d_k, \sigma) + \epsilon_k. \quad (12.2.14)$$

Step 3. $x_{k+1} = x_k + \alpha_k d_k$;
Evaluate $f(x_{k+1}), g_{k+1}, c(x_{k+1}), A_{k+1}$;

Step 4. *Compute* $\lambda_{k+1} = -(A_{k+1}^T A_{k+1})^{-1} A_{k+1}^T g_{k+1}$;
Set $s_k = \alpha d_k$, $y_k = \nabla_x L(x_{k+1}, \lambda_{k+1}) - \nabla_x L(x_k, \lambda_{k+1})$;
generate B_{k+1} *by updating* B_k *using a quasi-Newton formula;*
$k := k + 1$; *go to Step 2.* □

In (12.2.14), the penalty function $P(x, \sigma)$ is the L_1 exact penalty function, ϵ_k is a sequence of nonnegative numbers satisfying

$$\sum_{k=1}^{\infty} \epsilon_k < +\infty. \quad (12.2.15)$$

The global convergence result of the above algorithm is as follows.

Theorem 12.2.3 *Assume that* $f(x)$ *and* $c_i(x)$ *are continuously differentiable, and that there exist constants* $m, M > 0$ *such that*

$$m\|d\|^2 \leq d^T B_k d \leq M\|d\|^2 \quad (12.2.16)$$

holds for all k *and* $d \in \Re^n$, *if* $\|\lambda_k\|_\infty \leq \sigma$ *for all* k, *then any accumulation point of* $\{x_k\}$ *generated by Algorithm 12.2.2 is a KKT point of (8.1.1)-(8.1.3).*

Proof. If the theorem is not true, there exists a subsequence of $\{x_k\}$ converging to $\bar{x}$ which is not a KKT point. Therefore there exists a subset K_0 having infinitely many elements such that

$$\lim_{\substack{k\in K_0 \\ k\to\infty}} x_k = \bar{x}. \tag{12.2.17}$$

Without loss of generality, we can assume that

$$\lim_{\substack{k\in K_0 \\ k\to\infty}} \lambda_k = \bar{\lambda}, \quad \lim_{\substack{k\in K_0 \\ k\to\infty}} B_k = \bar{B}. \tag{12.2.18}$$

If

$$\lim_{\substack{k\in K_0 \\ k\to\infty}} \|d_k\| = 0, \tag{12.2.19}$$

from the relation

$$g_k + B_k d_k = A(x_k)\lambda_k, \tag{12.2.20}$$

it follows that

$$g(\bar{x}) = A(\bar{x})\bar{\lambda}. \tag{12.2.21}$$

This contradicts the fact that $\bar{x}$ is not a KKT point. Therefore we can assume that

$$\|d_k\| \geq \eta > 0, \quad \forall k \in K_0, \tag{12.2.22}$$

where η is a constant. The above relation and (12.2.13) imply that

$$P'_\alpha(x_k + \alpha d_k, \sigma)|_{\alpha=0} \leq -m\eta\|d_k\|, \tag{12.2.23}$$

holds for all $k \in K_0$. It follows from (12.2.23) and the continuity assumptions on the functions that there exists a positive constant $\bar{\eta}$ such that

$$\min_{0\leq\alpha\leq\delta} P(x_k + \alpha d_k, \sigma) \leq P(x_k, \sigma) - \bar{\eta} \tag{12.2.24}$$

hold for all $k \in K_0$. Thus,

$$P(x_{k+1}, \sigma) \leq P(x_k, \sigma) - \bar{\eta} + \epsilon_k, \quad \forall k \in K_0. \tag{12.2.25}$$

Consequently we can derive the inequality

$$\begin{aligned} \sum_{k\in K_0} \bar{\eta} &\leq \sum_{k\in K_0} [P(x_k,\sigma) - P(x_{k+1},\sigma)] + \sum_{k\in K_0} \epsilon_k \\ &\leq \sum_{k=1}^{\infty} [P(x_k,\sigma) - P(x_{k+1},\sigma)] + \sum_{k=1}^{\infty} \epsilon_k. \end{aligned} \tag{12.2.26}$$

Because $\lim_{k\to\infty} P(x_k,\sigma) = P(\bar{x},\sigma)$, it follows that

$$\sum_{k\in K_0} \bar{\eta} \leq P(x_1,\sigma) - P(\bar{x},\sigma) + \sum_{k=1}^{\infty} \epsilon_k < +\infty. \tag{12.2.27}$$

From the above inequality and $\bar{\eta} > 0$ we see that K_0 can have only finitely many elements, which contradicts our assumption in the beginning of the proof. This indicates that the theorem is true. $\Box$

The global convergence requires that

$$\sigma > \|\lambda_k\|_\infty \tag{12.2.28}$$

for all k. However, in practice it is very difficult to choose such a penalty parameter. If σ is too small, condition (12.2.28) may be violated. If σ is too large, the step-length α_k may tend to be too small to prevent the fast convergence of the algorithm. Powell [268] suggests using exact penalty function

$$P(x,\sigma_k) = f(x) + \sum_{i=1}^{m} (\sigma_k)_i |c_i^{(-)}(x)| \tag{12.2.29}$$

in the k-th iteration, where $(\sigma_k)_i > 0$ and these parameters are updated in the following way.

$$(\sigma_1)_i = (\lambda_1)_i, \tag{12.2.30}$$

$$(\sigma_k)_i = \max\left\{ |[\lambda_k]_i|, \frac{1}{2}[(\sigma_{k-1})_i + |(\lambda_k)_i|] \right\}, \quad k > 1, \tag{12.2.31}$$

for all $i = 1, \cdots, m$. The parameters σ_k defined above satisfy

$$(\sigma_k)_i \geq |(\lambda_k)_i|, \quad i = 1, 2, \cdots, m. \tag{12.2.32}$$

This very clever update technique allows the penalty parameters to change from iteration to iteration, and, intuitively, the inequality (12.2.32) offers a similar property to (12.2.28). But, because $(\sigma_k)_i$ are not constants, the conditions of Theorem 12.2.3 do not hold. And Chamberlain [53] gives an example to show that cycles may happen due to this update technique.

Now we discuss the update of B_{k+1}, which is usually generated by a certain quasi-Newton formula. From our analyses in Section 12.1, we hope that B_{k+1} is an approximation to the Hessian matrix of the Lagrangian function.

Similar to unconstrained optimization, we can apply the standard quasi-Newton updates using

$$
\begin{aligned}
s_k &= x_{k+1} - x_k, && (12.2.33) \\
y_k &= \nabla f(x_{k+1}) - \nabla f(x_k) \\
&\quad - \sum_{i=1}^{m} (\lambda_k)_i [\nabla c_i(x_{k+1}) - \nabla c_i(x_k)]. && (12.2.34)
\end{aligned}
$$

A crucial difference is that line

$$
s_k^T y_k > 0, \tag{12.2.35}
$$

which would be always true for unconstrained optimization. Therefore, for example, we can not directly apply the BFGS update. Powell [268] suggests that y_k be replaced by

$$
\bar{y}_k = \begin{cases} y_k, & \text{if } s_k^T y_k \geq 0.2 s_k^T B_k s_k, \\ \theta_k y_k + (1 - \theta_k) B_k s_k, & \text{otherwise} \end{cases} \tag{12.2.36}
$$

where

$$
\theta_k = \frac{0.8 s_k^T B_k s_k}{s_k^T B_k s_k - s_k^T y_k}. \tag{12.2.37}
$$

The vector $\bar{y}_k$ defined above satisfies $s_k^T \bar{y}_k > 0$.

The idea of such a choice of $\bar{y}_k$ is to obtain an update vector using the convex combination of y_k and $B_k s_k$. Because $B_k s_k$ can also be viewed as an approximation to y_k, because it satisfies (assuming that B_k is positive definite)

$$
s_k^T (B_k s_k) > 0, \tag{12.2.38}
$$

it is very natural to use the convex combination of y_k and $B_k s_k$. The geometric interpretation of Powell's formula is as follows. Suppose we normalize the length of the projection of $B_k s_k$ to direction s_k. The rule (12.2.36)-(12.2.37) is in fact to choose $\bar{y}_k$ from the line segment between y_k and $B_k s_k$ that is as close to y_k as possible and whose projection to s_k is at least 0.2. This is shown in Figure 12.2.1.

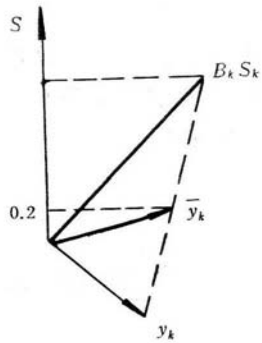

Figure 12.2.1

Having computed the vector $\bar{y}_k$, we can now apply the BFGS formula to update B_{k+1}:

$$B_{k+1} = B_k - \frac{B_k s_k s_k^T B_k^T}{s_k^T B_k s_k} + \frac{\bar{y}_k \bar{y}_k^T}{s_k^T \bar{y}_k}. \tag{12.2.39}$$

Another way to modify y_k is to use

$$\hat{y}_k = y_k + 2\rho \sum_{i=1}^{m} -c_i(x_k)\nabla c_i(x_k) \tag{12.2.40}$$

to replace y_k, where $\rho > 0$ is a parameter. Because

$$\hat{y}_k \approx [\nabla^2 L(x_k, \lambda_k) + 2\rho A(x_k)A(x_k)^T] s_k, \tag{12.2.41}$$

updating B_k by using $\hat{y}_k$ can be viewed as making B_{k+1} approximate the Hessian matrix of the augmented Lagrange function. An advantage of this choice is that

$$s_k^T \hat{y}_k > 0 \tag{12.2.42}$$

can usually be satisfied. If $s_k^T \hat{y}_k \leq 0$, we can always make (12.2.42) hold by increasing ρ, unless $\|A(x_k)s_k\| = 0$. Normally, the Hessian matrix of the augmented Lagrange function is positive definite, thus it is very reasonable to use a positive definite matrix B_k to approximate it.

12.3 Superlinear Convergence of SQP Step

In order to prove the superlinear convergence property of the sequential quadratic programming method, i.e.

$$\lim_{k\to\infty} \frac{\|x_{k+1} - x^*\|}{\|x_k - x^*\|} = 0, \tag{12.3.1}$$

we only need to show that the search direction d_k satisfies

$$\lim_{k\to\infty} \frac{\|x_k + d_k - x^*\|}{\|x_k - x^*\|} = 0 \tag{12.3.2}$$

and that the line search condition will allow $\alpha_k = 1$ for all large k if (12.3.2) holds. Thus, the important thing is to show that the search direction generated by the SQP method satisfies (12.3.2). A step d_k satisfying (12.3.2) is called a superlinearly convergent step. In this section, we discuss the conditions for ensuring the sequential quadratic programming method to produce superlinearly convergent steps.

Throughout this section, we make the following assumptions.

Assumption 12.3.1

1) $f(x), c_i(x)$ are twice continuously differentiable;

2) $x_k \to x^*$;

3) x^* is a KKT point and

$$\nabla c_i(x^*), \quad i \in E \cup I(x^*) \tag{12.3.3}$$

are linearly independent. Let $A(x^*)$ be the $n \times |E \cup I(x^*)|$ matrix consisting of the vectors given in (12.3.3). For all nonzero vectors d satisfying

$$A(x^*)^T d = 0, \tag{12.3.4}$$

we have that

$$d^T W(x^*, \lambda^*) d \neq 0, \tag{12.3.5}$$

where $W(x^*, \lambda^*)$ is defined by (12.1.8), and λ^* is the Lagrange multiplier at x^*.

The above assumptions are often used for local convergence analyses of algorithms for constrained optimization. For example, relations (12.3.4) and (12.3.5) hold if we assume the second-order sufficient condition

$$d^T W(x^*, \lambda^*) d > 0, \quad \forall d \neq 0,\ A(x^*)^T d = 0. \tag{12.3.6}$$

We also make the assumption that the active set at the solution can be identified when the iterations are very close to the solution. Therefore, when k is sufficiently large, the search direction d_k is actually the solution of an equality constrained quadratic programming subproblem.

Assumption 12.3.2 *For sufficiently large k, d_k is a solution of*

$$\min_{d \in \Re^n} \quad g_k^T d + \frac{1}{2} d^T B_k d \tag{12.3.7}$$

$$s.t. \quad c_i(x_k) + d^T \nabla c_i(x_k) = 0, \quad i \in E \cup I(x^*). \tag{12.3.8}$$

Under Assumption 12.3.2, for all large k there exists $\lambda_k \in \Re^{|E \cup I(x^*)|}$ such that

$$g_k + B_k d_k = A(x_k)\lambda_k, \tag{12.3.9}$$

$$A(x_k)^T d_k = -\hat{c}(x_k), \tag{12.3.10}$$

where $\hat{c}(x)$ is a vector whose elements are $c_i(x)(i \in E \cup I(x^*))$.

Theorem 12.3.3 *Under the conditions of Assumptions 12.3.1 and 12.3.2, d_k is a superlinearly convergent step, namely*

$$\lim_{k \to \infty} \frac{\|x_k + d_k - x^*\|}{\|x_k - x^*\|} = 0 \tag{12.3.11}$$

if and only if

$$\lim_{k \to \infty} \frac{\|P_k(B_k - W(x^*, \lambda^*))d_k\|}{\|d_k\|} = 0, \tag{12.3.12}$$

where P_k is a projection from $\Re^n$ onto the null space of $A(x_k)^T$:

$$P_k = I - A(x_k)(A(x_k)^T A(x_k))^{-1} A(x_k)^T. \tag{12.3.13}$$

Proof. From (12.3.9) and the definition of P_k, we have that

$$\begin{aligned} P_kB_kd_k &= -P_kg_k = -P_k[\nabla f(x_k) - A(x_k)\lambda^*] \\ &= -P_k[\nabla_x L(x_k,\lambda^*) - \nabla_x L(x^*,\lambda^*)] \\ &= -P_kW(x^*,\lambda^*)(x_k - x^*) + O(\|x_k - x^*\|^2). \end{aligned} \tag{12.3.14}$$

Therefore, it follows that

$$\begin{aligned} P_k(B_k - W(x^*,\lambda^*))d_k &= -P_kW(x^*,\lambda^*)[x_k + d_k - x^*] \\ &\quad +O(\|x_k - x^*\|^2). \end{aligned} \tag{12.3.15}$$

Using relation (12.3.10) and

$$\begin{aligned} \hat{c}(x_k) &= \hat{c}(x_k) - \hat{c}(x^*) \\ &= A(x_k)^T(x_k - x^*) + O(\|x_k - x^*\|^2), \end{aligned} \tag{12.3.16}$$

we can show that

$$A(x_k)^T(x_k + d_k - x^*) = O(\|x_k - x^*\|^2). \tag{12.3.17}$$

Equations (12.3.15) and (12.3.17) can be rewritten in matrix form:

$$\begin{aligned} \begin{bmatrix} P_kW(x^*,\lambda^*) \\ A(x_k)^T \end{bmatrix}(x_k + d_k - x^*) &= \begin{bmatrix} -P_k(B_k - W(x^*,\lambda^*))d_k \\ 0 \end{bmatrix} \\ &+ O(\|x_k - x^*\|^2). \end{aligned} \tag{12.3.18}$$

Define the matrix

$$G^* = \begin{bmatrix} P_*W(x^*,\lambda^*) \\ A(x^*)^T \end{bmatrix}, \tag{12.3.19}$$

where $P_* = I - A(x^*)(A(x^*)^TA(x^*))^{-1}A(x^*)^T$. For any $d \in \Re^n$, if $G^*d = 0$ we have that

$$A(x^*)^Td = 0, \tag{12.3.20}$$

$$d^TP_*W(x^*,\lambda^*)d = 0. \tag{12.3.21}$$

From (12.3.20) it follows that $P_*d = d$. Thus,

$$d^TW(x^*,\lambda^*)d = 0. \tag{12.3.22}$$

The above relation and Assumption 12.3.1 show that $d = 0$. Therefore matrix G^* is a full column rank matrix. Hence, from (12.3.18) and the fact that $x_k \to x^*$ we can see that (12.3.11) is equivalent to

$$\lim_{k\to\infty} \frac{\|P_k(B_k - W(x^*, \lambda^*))d_k\|}{\|x_k - x^*\|} = 0. \tag{12.3.23}$$

Using the equivalence between (12.3.23) and (12.3.11) and that between (12.3.11) and

$$\lim_{k\to\infty} \|x_k - x^*\| \big/ \|d_k\| = 1, \tag{12.3.24}$$

we can show that (12.3.23) is equivalent to (12.3.12). This completes the proof. $\square$

Using relation (12.3.9) and $\lambda_k \to \lambda^*$, we have that

$$\begin{aligned} & W(x^*, \lambda^*)d_k = W(x_k, \lambda_k)d_k + o(\|d_k\|) \\ = \; & \nabla f(x_k + d_k) - A(x_k + d_k)\lambda_k - \nabla f(x_k) + A(x_k)\lambda_k + o(\|d_k\|) \\ = \; & \nabla f(x_k + d_k) - A(x_k + d_k)\lambda_k + B_k d_k + o(\|d_k\|). \end{aligned} \tag{12.3.25}$$

Therefore,

$$\begin{aligned} P_k(B_k - W(x^*, \lambda^*))d_k \;\; = \;\; & -P_k[\nabla f(x_k + d_k) - A(x_k + d_k)\lambda_k] \\ & + o(\|d_k\|). \end{aligned} \tag{12.3.26}$$

From the above relation and Theorem 12.3.3, we can get the following result.

Corollary 12.3.4 *Under the assumptions of Theorem 12.3.3, (12.3.11) is equivalent to*

$$\lim_{k\to\infty} \frac{\|P_k[\nabla f(x_k + d_k) - A(x_k + d_k)\lambda_k\|}{\|d_k\|} = 0. \tag{12.3.27}$$

From Theorem 12.3.3, we should choose B_k such that (12.3.12) is satisfied in order to have superlinear convergence, namely B_k should be a good approximation to $W(x^*, \lambda^*)$.

12.4 Maratos Effect

For unconstrained optimization, if x^* is a stationary point at which the second-order sufficient condition holds, namely

$$\nabla^2 f(x^*) \quad \text{positive definite,} \tag{12.4.1}$$

if $x_k \to x^*$, and if d_k is a superlinearly convergent step, then

$$f(x_k + d_k) < f(x_k) \tag{12.4.2}$$

holds for all large k. That is to say, superlinearly convergent steps are acceptable for unconstrained problems. However, this is not always true for constrained problems. Such a phenomenon was first discovered by Maratos [209], so it is called the Maratos Effect.

Consider the equality constrained optimization problem

$$\min_{x=(u,v)\in\Re^2} \quad f(x) = 3v^2 - 2u, \tag{12.4.3}$$

$$\text{s.t.} \quad c(x) = u - v^2 = 0. \tag{12.4.4}$$

It is easy to see that $x^* = (0,0)^T$ is the unique minimizer and condition 3) of Assumption 12.3.1 is satisfied. In fact, the second-order sufficient condition holds at x^*. Consider any points that are close to the solution x^* and that have the form

$$\bar{x}(\epsilon) = (u(\epsilon), v(\epsilon))^T = (\epsilon^2, \epsilon)^T \tag{12.4.5}$$

where $\epsilon > 0$ is a small parameter. Let $B = W(x^*, \lambda^*)$; the quadratic programming subproblem is

$$\min_{d\in\Re^2} \quad d^T \begin{pmatrix} -2 \\ 6\epsilon \end{pmatrix} + \frac{1}{2} d^T \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix} d, \tag{12.4.6}$$

$$\text{s.t.} \quad d^T \begin{pmatrix} 1 \\ -2\epsilon \end{pmatrix} = 0. \tag{12.4.7}$$

It is easy to see that the solution of (12.4.6)-(12.4.7) is

$$\bar{d}(\epsilon) = \begin{bmatrix} -2\epsilon^2 \\ -\epsilon \end{bmatrix}. \tag{12.4.8}$$

Therefore, we have that

$$\|\bar{x}(\epsilon) + \bar{d}(\epsilon) - x^*\| = O(\|\bar{x}(\epsilon) - x^*\|^2). \tag{12.4.9}$$

Thus, $\bar{d}(\epsilon)$ is a superlinearly convergent step. Direct calculation indicates that

$$f(\bar{x}(\epsilon) + \bar{d}(\epsilon)) = 2\epsilon^2, \tag{12.4.10}$$
$$c(\bar{x}(\epsilon) + \bar{d}(\epsilon)) = -\epsilon^2. \tag{12.4.11}$$

Because

$$f(\bar{x}(\epsilon)) = \epsilon^2, \tag{12.4.12}$$
$$c(\bar{x}(\epsilon)) = 0, \tag{12.4.13}$$

we have that

$$f(\bar{x}(\epsilon) + \bar{d}(\epsilon)) > f(\bar{x}(\epsilon)), \tag{12.4.14}$$
$$|c(\bar{x}(\epsilon) + \bar{d}(\epsilon))| > |c(\bar{x}(\epsilon))|. \tag{12.4.15}$$

This example shows that even though $\bar{d}(\epsilon)$ is a superlinearly convergent step (namely $\bar{x}(\epsilon) + \bar{d}(\epsilon)$ is much closer to x^* than $\bar{x}(\epsilon)$), the point $\bar{x}(\epsilon) + \bar{d}(\epsilon)$ is "worse" than $\bar{x}(\epsilon)$ from the objective function values and from the constraint violations. In fact, for any penalty functions $P_{\sigma,h}(x)$ having the form of (10.6.2), we would have that

$$P_{\sigma,h}(\bar{x}(\epsilon) + \bar{d}(\epsilon)) > P_{\sigma,h}(\bar{x}(\epsilon)). \tag{12.4.16}$$

Especially, when the merit function is the L_1 exact penalty function, $\bar{x}(\epsilon) + \bar{d}(\epsilon)$ is not acceptable.

The Maratos Effect shows that for many penalty functions a superlinearly convergent step may not be accepted, which, sometimes, prevents the algorithm from fast convergence.

There are mainly three ways to overcome the Maratos Effect. The first one is to relax the line search conditions. Roughly speaking, since the search direction d_k is a superlinearly convergent step, we should choose $\alpha_k = 1$ as often as possible provided that convergence is ensured. The second one is to use a second-order correction step $\hat{d}_k$, where $\hat{d}_k$ satisfies $\|\hat{d}_k\| = O(\|d_k\|^2)$, and $P_\sigma(x_k + d_k + \hat{d}_k) < P_\sigma(x_k)$. In this way, $d_k + \hat{d}_k$ is an acceptable step and it is still a superlinearly convergent step. The third way is to use smooth exact penalty functions as merit functions. If the penalty function $P_\sigma(x)$ is smooth, we can show that

$$P_\sigma(x_k + d_k) < P_\sigma(x_k) \tag{12.4.17}$$

for all large k as long as (12.3.11) holds.

We will discuss these three techniques in the following sections.

12.5 Watchdog Technique

The nature of the Maratos Effect is the inequality

$$P_\sigma(x_k + d_k) > P_\sigma(x_k), \tag{12.5.1}$$

makes $x_{k+1} \neq x_k + d_k$, therefore the superlinearly convergent property is destroyed. In the Watchdog technique proposed by Chamberlain et. al. [54], the standard linear search which implies that

$$P_\sigma(x_{k+1}) < P_\sigma(x_k), \tag{12.5.2}$$

is used in some iterations, but in the other iterations, the line search conditions are relaxed. The relaxed line search can be either simply $\alpha_k = 1$ or requiring the Lagrange function to be reduced. Assume that the new point obtained in one iteration yields a sufficient reduction on the merit function $P_\sigma(x)$; comparing with the best point in the previous iterations, we can use the relaxed line search in the next iteration.

Define the function

$$P_\sigma(x) = f(x) + \sum_{i=1}^{m_e} \sigma_i |c_i(x)| + \sum_{i=m_e+1}^{m} \sigma_i |\min[0, c_i(x)]|, \tag{12.5.3}$$

and the approximate models

$$\begin{aligned} P_\sigma^{(k)}(x) &= f(x_k) + (x - x_k)^T \nabla f(x_k) + \frac{1}{2}(x - x_k)^T B_k (x - x_k) \\ &+ \sum_{i=1}^{m_e} \sigma_i |c_i(x_k) + (x - x_k)^T \nabla c_i(x_k)| \\ &+ \sum_{i=m_e+1}^{m} \sigma_i |\min[0, c_i(x_k) + (x - x_k)^T \nabla c_i(x_k)]|. \end{aligned} \tag{12.5.4}$$

Assume that $l \leq k$ is the index in which the best point has been found up to the k-th iteration, namely

$$P_\sigma(x_l) = \min_{1 \leq i \leq k} P_\sigma(x_i). \tag{12.5.5}$$

Let $\beta \in \left(0, \frac{1}{2}\right)$ be a given positive constant. If the iterate point obtained in the k-th iteration $x_{k+1} = x_k + \alpha_k d_k$ satisfies

$$P_\sigma(x_{k+1}) \leq P_\sigma(x_l) - \beta[P_\sigma(x_l) - P_\sigma^{(l)}(x_{l+1})], \tag{12.5.6}$$

then we say x_{k+1} (comparing to x_l) yields a "sufficient" reduction on the merit function $P_\sigma(x)$.

The following is an algorithm with the Watchdog technique.

Algorithm 12.5.1 *(Watchdog Method)*

Step 1. *Given $x_1 \in \Re^n$, a positive constant $\bar{n}$.*
Set line search type to be standard; $k := l := 1$;

Step 2. *Compute the search direction d_k;*
Carry out line search using the line search type, obtaining $\alpha_k > 0$;
$x_{k+1} = x_k + \alpha_k d_k$;

Step 3. *if (12.5.6) holds, then set the next line search type to be "relaxed", otherwise to be standard.*

Step 4. *if $P_\sigma(x_{k+1}) \leq P_\sigma(x_l)$, then $l := k + 1$;*

Step 5. *if $k < l + \bar{n}$, then go to Step 6;*
$x_{k+1} := x_l$; $l := k + 1$;

Step 6. *if convergence criterion is satisfied then stop;*
$k := k + 1$; go to Step 2. □

Actually, if the "relaxed" line search conditions are the same as the standard condition, the above algorithm is the original method that is based on the standard line searches. Therefore, the Watchdog method is a generalization of the standard method.

Assume the standard line search condition is

$$P_\sigma(x_{k+1}) \leq P_\sigma(x_k) - \beta[P_\sigma(x_k) - P_\sigma^{(k)}(x_{k+1})]. \tag{12.5.7}$$

From the descriptions of the above algorithm, we know that there exists $k \le l + \bar{n} + 1$ such that

$$P_\sigma(x_{k+1}) \le P_\sigma(x_l) - \beta[P_\sigma(x_l) - P_\sigma^{(l)}(x_{l+1})]. \tag{12.5.8}$$

Thus, the watchdog method will reduce the merit function $P_\sigma(x)$ in every $\bar{n}+1$ iterations, even though it can not guarantee the monotonically decreasing of $P_\sigma(x_k)$. Let $l(j)$ be the j-th value of l; from the discussions above we see that

$$l(j) < l(j+1) \le l(j) + \bar{n} + 2. \tag{12.5.9}$$

If we assume that the sequence $\{x_k\}$ is bounded, then $P_\sigma(x_{l(j)})$ will not tend to negative infinity. Thus, it follows from the inequality

$$P_\sigma(x_{l(j+1)}) \le P_\sigma(x_{l(j)}) - \beta[P_\sigma(x_{l(j)}) - P_\sigma^{(l(j))}(x_{l(j)+1})] \tag{12.5.10}$$

that

$$\sum_{j=1}^{\infty}[P_\sigma(x_{l(j)}) - P_\sigma^{(l(j))}(x_{l(j)+1})] < +\infty. \tag{12.5.11}$$

The above relation shows that there exists an accumulation point of $\{x_k\}$ that is a KKT point of the constrained optimization problem.

12.6 Second-Order Correction Step

A second-order correction step is a vector $\hat{d}_k$ such that

$$\|\hat{d}_k\| = O(\|d_k\|^2) \tag{12.6.1}$$

and

$$P_\sigma(x_k + d_k + \hat{d}_k) < P_\sigma(x_k) \tag{12.6.2}$$

for all sufficiently large k. Consider that $\hat{d}_k$ is defined as a solution of the following quadratic programming problem:

$$\min_{d \in \Re^n} \quad g_k^T(d_k + d) + \frac{1}{2}(d_k + d)^T B_k (d_k + d), \tag{12.6.3}$$

$$\text{s.t.} \quad c_i(x_k + d_k) + a_i(x_k)^T d = 0, \quad i \in E, \tag{12.6.4}$$

$$c_i(x_k + d_k) + a_i(x_k)^T d \ge 0, \quad i \in I, \tag{12.6.5}$$

where d_k is the solution of (12.2.1)-(12.2.3).

For simplicity, we assume that all the constraints are equality constraints. We also assume that the second-order sufficient conditions hold at x^* and that $x_k \to x^*$. From the KKT condition there exist $\lambda_k \in \Re^m$ and $\hat{\lambda}_k \in \Re^m$ such that

$$B_k d_k = -g_k + A(x_k)\lambda_k, \tag{12.6.6}$$
$$A(x_k)^T d_k = -c(x_k) \tag{12.6.7}$$

and that

$$\begin{aligned} B_k d_k + B_k \hat{d}_k &= -g_k + A(x_k)\hat{\lambda}_k, && (12.6.8) \\ A(x_k)^T \hat{d}_k &= -c(x_k + d_k). && (12.6.9) \end{aligned}$$

From (12.6.6) and (12.6.8) we see that

$$P_k B_k \hat{d}_k = 0, \tag{12.6.10}$$

where P_k is defined by (12.3.13). We make the following assumptions.

Assumption 12.6.1

1) $x_k \to x^*$*;*

2) $A(x^*)$ *is full column rank;*

3) there exist positive constants $\bar{m}$ *and* $\bar{M}$ *such that* $\|B_k\| \le \bar{M}$ *and that*

$$d^T B_k d \ge \bar{m}\|d\|_2^2 \tag{12.6.11}$$

holds for all d *satisfying* $A(x_k)^T d = 0$ *for all* k*.*

From the above assumptions we can show the following lemma.

Lemma 12.6.2 *Under the conditions of Assumption 12.6.1, there exists a positive constant* η *such that*

$$\left\| \begin{pmatrix} P_k B_k \\ A(x_k)^T \end{pmatrix} d \right\|_2 \ge \eta \|d\|_2 \tag{12.6.12}$$

holds for all $d \in \Re^n$ *and all sufficiently large* k*.*

Proof. Let the QR factorization of $A(x_k)$ be

$$A(x_k) = [Y_k \ Z_k] \begin{bmatrix} R_k \\ 0 \end{bmatrix}. \tag{12.6.13}$$

Because $A(x^*)$ is nonsingular, there exists k_0 such that for $k \geq k_0$ we have

$$\|R_k^{-1}\|_2 \leq \hat{\eta}, \tag{12.6.14}$$

where $\hat{\eta} > 0$ is a constant. Therefore,

$$\|A(x_k)^T d\|_2 = \|R_k^T Y_k^T d\|_2 \geq \frac{1}{\hat{\eta}} \|Y_k^T d\|_2, \tag{12.6.15}$$

for $k \geq k_0$. Using the relation $Y_k Y_k^T + Z_k Z_k^T = I$, we can show that

$$\begin{aligned} \|P_k B_k d\|_2 &= \|Z_k Z_k^T B_k d\|_2 \\ &= \|Z_k Z_k^T B_k Y_k Y_k^T d + Z_k Z_k^T B_k Z_k Z_k^T d\|_2 \\ &\geq \|Z_k Z_k^T B_k Z_k Z_k^T d\|_2 - \|B_k\|_2 \|Y_k^T d\|_2 \\ &\geq \bar{m} \|Z_k^T d\|_2 - \bar{M} \|Y_k^T d\|_2. \end{aligned} \tag{12.6.16}$$

Thus, if

$$\|Y_k^T d\| \geq \frac{\bar{m}}{2\bar{M}} \|Z_k^T d\|, \tag{12.6.17}$$

it follows from (12.6.15) that

$$\begin{aligned} \|A(x_k)^T d\|_2 &\geq \frac{1}{\hat{\eta}} \|Y_k^T d\|_2 \\ &\geq \frac{\frac{\bar{m}}{2\bar{M}}}{\hat{\eta}\sqrt{1 + \left(\frac{\bar{m}}{2\bar{M}}\right)^2}} \|d\|_2. \end{aligned} \tag{12.6.18}$$

If (12.6.17) does not hold, it follows from (12.6.16) that

$$\|P_k B_k d\|_2 \geq \frac{1}{2} \bar{m} \|Z^T d\|_2 \geq \frac{\bar{M}}{\sqrt{1 + \left(\frac{2\bar{M}}{\bar{m}}\right)^2}} \|d\|_2. \tag{12.6.19}$$

Therefore, when $k \geq k_0$, either of (12.6.18) and (12.6.19) must hold. Let

$$\eta = \min\left\{\frac{1}{\hat{\eta}}, \bar{M}\right\} \frac{1}{\sqrt{1 + 4(\bar{M}/\bar{m})^2}}, \tag{12.6.20}$$

then we see that (12.6.12) holds for all $k \geq k_0$ and all $d \in \Re^n$. $\square$

Using (12.6.9)-(12.6.10), we have that

$$\left[\begin{array}{c} P_k B_k \\ A(x_k)^T \end{array} \right] \hat{d}_k = \left[\begin{array}{c} 0 \\ -c(x_k + d_k) \end{array} \right] = O(\|d_k\|_2^2). \tag{12.6.21}$$

Now, from the above relation and Lemma 12.6.2 we can show the following lemma.

Lemma 12.6.3 *Under the conditions of Assumption 12.6.1, there exists a positive constant* $\bar{\eta} > 0$ *such that*

$$\|\hat{d}_k\|_2 \leq \bar{\eta} \|d_k\|_2^2. \tag{12.6.22}$$

To this end, we have shown that the step defined by (12.6.3)-(12.6.5) is indeed a second-order correction step.

In the following we show that the second-order correction step $\hat{d}_k$ will make the step $d_k + \hat{d}_k$ acceptable. First, using (12.6.9) we see that

$$\begin{aligned} c(x_k + d_k + \hat{d}_k) &= c(x_k + d_k) + A(x_k)^T \hat{d}_k + o(\|\hat{d}_k\|) \\ &= o(\|d_k\|^2) = o(\|x_k - x^*\|^2). \end{aligned} \tag{12.6.23}$$

Define the vector

$$\bar{d}_k = -(A(x_k)^T)^+ c(x_k + d_k) - P_k(x_k + d_k - x^*), \tag{12.6.24}$$

then it follows that

$$\begin{aligned} \|x_k + d_k + \bar{d}_k &- x^*\| = \|(I - P_k)(x_k + d_k - x^*) \\ &\quad -(A(x_k)^T)^+ c(x_k + d_k)\| \\ &= \|(I - P_k)(x_k + d_k - x^*) \\ &\quad -(A(x_k)^T)^+ A(x_k)^T (x_k + d_k - x^*)\| \\ &\quad +o(\|x_k - x^*\|^2) = o(\|x_k - x^*\|^2). \end{aligned} \tag{12.6.25}$$

Furthermore, it follows from (12.6.24) that

$$A(x_k)^T \bar{d}_k = -c(x_k + d_k). \tag{12.6.26}$$

If we assume not only (12.3.12) but also

$$\frac{\|(B_k - W(x^*, \lambda^*))d\|}{\|d\|} \rightarrow 0 \tag{12.6.27}$$

holds for $d = d_k + \hat{d}_k$, and $d = d_k + \bar{d}$, then it follows that

$$\begin{aligned} & (g_k - A_k\lambda^*)^T d + \frac{1}{2} d^T B_k d \\ = \ & L(x_k + d, \lambda^*) - L(x_k, \lambda^*) + o(\|d\|^2) + o(\|x_k - x^*\|^2) \\ = \ & L(x_k + d, \lambda^*) - L(x_k, \lambda^*) + o(\|x_k - x^*\|^2) \end{aligned} \tag{12.6.28}$$

holds for $d = d_k + \hat{d}_k$ and $d = d_k + \bar{d}_k$. From the definition of $\hat{d}_k$, we can show that

$$\begin{aligned} g_k^T \hat{d}_k \ &+ \ \frac{1}{2}(d_k + \hat{d}_k)^T B_k (d_k + \hat{d}_k) \\ &\le \ g_k^T \bar{d}_k + \frac{1}{2}(d_k + \bar{d}_k)^T B_k (d_k + \bar{d}_k). \end{aligned} \tag{12.6.29}$$

If follows from (12.6.28) and (12.6.29) that

$$\begin{aligned} L(x_k + d_k + \hat{d}_k, \lambda^*) \ &\le \ L(x_k + d_k + \bar{d}_k, \lambda^*) + o(\|x_k - x^*\|^2) \\ &\le \ L(x^*, \lambda^*) + o(\|x_k - x^*\|^2). \end{aligned} \tag{12.6.30}$$

The above inequality and (12.6.23) imply that

$$f(x_k + d_k + \hat{d}_k) \le f(x^*) + o(\|x_k - x^*\|^2). \tag{12.6.31}$$

It follows from the above relation and (12.6.23) that

$$P_\sigma(x_k + d_k + \hat{d}_k) \le P_\sigma(x^*) + o(\|x_k - x^*\|^2). \tag{12.6.32}$$

Under the second-order sufficient condition, there exists a positive constant $\delta > 0$ such that

$$P_\sigma(x_k) \ge P_\sigma(x^*) + \delta\|x_k - x^*\|^2. \tag{12.6.33}$$

Therefore, by the above two inequalities we can deduce that

$$P_\sigma(x_k + d_k + \hat{d}_k) < P_\sigma(x_k). \tag{12.6.34}$$

To be more exact, from (12.6.32)-(12.6.33) we can show that

$$\lim_{k\to\infty} \frac{P_\sigma(x_k) - P_\sigma(x_k + d_k + \hat{d}_k)}{P_\sigma(x_k) - P_\sigma(x^*)} = 1. \tag{12.6.35}$$

Therefore, it follows from (12.6.22) and (12.6.34), that

$$\lim_{k\to\infty} \frac{\|x_k + d_k + \hat{d}_k - x^*\|}{\|x_k - x^*\|} = 0, \tag{12.6.36}$$

namely, $d_k + \hat{d}_k$ is a superlinearly convergent step and it is acceptable.

Another way to compute a second-order correction step is to solve the following subproblem:

$$\min_{d\in\Re^n} \quad \tilde{g}_k^T d + \frac{1}{2} d^T B_k d, \tag{12.6.37}$$

$$\text{s.t.} \quad c_i(x_k) + a_i(x_k)^T d = 0, \quad i \in E, \tag{12.6.38}$$

$$c_i(x_k) + a_i(x_k)^T d \geq 0, \quad i \in I, \tag{12.6.39}$$

where

$$\tilde{g}_k = g_k + \frac{1}{2}\sum_{i=1}^{m}(\lambda_k)_i[\nabla c_i(x_k) - \nabla c_i(x_k + d_k)], \tag{12.6.40}$$

and where λ_k is the Lagrange multiplier of the quadratic programming subproblem (12.2.1)-(12.2.3). It can be shown that the search direction defined by (12.6.37)-(12.6.39) is a superlinearly convergent step and is also an acceptable step. For more detailed discussions, please see Mayne and Polak [214] and Fukushima [142].

12.7 Smooth Exact Penalty Functions

The reason for the Maratos Effect to happen is because the merit function used to carry out line search is nonsmooth. If $P(x)$ is a smooth function, if x^* is its minimizer, and if $\nabla^2 P(x^*)$ is positive definite, we can easily see that, for all x sufficiently close to x^*,

$$\bar{M}\|x - x^*\|^2 \geq P(x) - P(x^*) \geq \bar{m}\|x - x^*\|^2, \tag{12.7.1}$$

where $\bar{M} \geq \bar{m}$ are two positive constants. Therefore if

$$\frac{\|x_k + d_k - x^*\|}{\|x_k - x^*\|} \to 0, \tag{12.7.2}$$

it is easy to show that

$$\begin{aligned} P(x_k + d_k) &\leq P(x^*) + \bar{M}\|x_k + d_k - x^*\|^2 \\ &< P(x^*) + \bar{m}\|x_k - x^*\|^2 \leq P(x_k) \end{aligned} \tag{12.7.3}$$

holds for sufficiently large k. Therefore, the Maratos Effect can be avoided if we use a smooth exact penalty function as the merit function.

Consider the equality constrained optimization problem:

$$\min_{x \in \Re^n} \quad f(x), \tag{12.7.4}$$

$$\text{s.t.} \quad c(x) = 0. \tag{12.7.5}$$

We use Fletcher's smooth exact penalty function (10.5.4) as the merit function. Because the derivative of function (10.5.4) needs to compute the second-order derivatives of $f(x)$ and $c(x)$, Powell and Yuan [277] uses an approximate form of (10.5.4):

$$\begin{aligned}\Phi_{k,i}(\alpha\beta_{k,i}) &= f(x_k + \alpha\beta_{k,i}d_k) \\ &- [\lambda(x_k) + \alpha(\lambda(x_k + \beta_{k,i}d_k) - \lambda(x_k))]^T c(x_k + \alpha\beta_{k,i}d_k) \\ &+ \frac{1}{2}\sigma_{k,i}\|c(x_k + \alpha\beta_{k,i}d_k)\|_2^2, \quad 0 \le \alpha \le 1,\end{aligned} \tag{12.7.6}$$

where d_k is a solution of the quadratic programming subproblem (12.2.1)-(12.2.3), $\beta_{k,i}$ is the $(i+1)$-th trial step length in the k-th iteration, and $\sigma_{k,i}$ is the current penalty parameter which satisfies that

$$\begin{aligned}\Phi'_{k,i}(0) &\le -\frac{1}{2}[d_k^T B_k d_k + \sigma_{k,i}\|c(x_k)\|_2^2] \\ &\le -\frac{1}{4}\sigma_{k,i}\|c(x_k)\|_2^2.\end{aligned} \tag{12.7.7}$$

The Powell and Yuan's method can be stated as follows:

Algorithm 12.7.1 *(Powell and Yuan's Method)*

Step 1. Given $x_1 \in \Re^n$, $\beta_1 \in (0,1)$, $\beta_2 \in (\beta_1, 1)$, $\mu \in (0, 1/2)$, $\sigma_{1,-1} > 0$, $B_1 \in \Re^{n\times n}$, $\epsilon \ge 0$. $k := 1$;

Step 2. Solve (12.2.1)-(12.2.3), giving d_k;
if $\|d_k\| \le \epsilon$ *then stop;*
let $i = 0$, $\beta_{k,0} = 1$;

Step 3. Choose $\sigma_{k,i}$ *such that (12.7.7) holds; if*

$$\Phi_{k,i}(\beta_{k,i}) \le \Phi_{k,i}(0) + \mu\beta_{k,i}\Phi'_{k,i}(0), \tag{12.7.8}$$

then go to Step 4.
$i := i+1$, $\beta_{k,i} \in [\beta_1, \beta_2]\beta_{k,i-1}$; *go to Step 3;*

Step 4. $x_{k+1} = x_k + \beta_{k,i} d_k$; $\sigma_{k+1,-1} = \sigma_{k,i}$; *update* B_{k+1};
$k := k+1$; *go to Step 2.*

For the above algorithm, it can be shown that the following lemma holds.

Lemma 12.7.2 *Assume that* $\{x_k\}$, $\{d_k\}$, $\{B_k\}$ *are bounded. If* $A(x) = \nabla c(x)^T$ *is full column rank for all* $x \in \Re^n$ *and if there exists a constant* $\delta > 0$ *such that*

$$d^T B_k d \geq \delta \|d\|_2^2, \quad \forall A(x_k)^T d = 0 \tag{12.7.9}$$

holds for all k, *then there exists a positive integer* k' *such that*

$$\sigma_{k,i} = \sigma_{k',0} = \bar{\sigma} > 0 \tag{12.7.10}$$

for all $k \geq k'$ *and that*

$$\lim_{k\to\infty} \|d_k\| = 0. \tag{12.7.11}$$

Using this lemma, we can prove the global convergence result of Algorithm 12.7.1

Theorem 12.7.3 *Under the conditions of Lemma 12.7.2, any accumulation point of* $\{x_k\}$ *generated by Algorithm 12.7.1 is a KKT point of (12.7.4)-(12.7.5).*

Now we show that when the iterates are close to a solution, any superlinearly convergent step will be accepted by Algorithm 12.7.1.

Lemma 12.7.4 *Suppose that the assumptions of Lemma 12.7.2 are satisfied, and assume that the sequence* $\{x_k\}$ *generated by Algorithm 12.7.1 converges to* x^*. *For any subsequence* $\{k_i, i = 1, 2, \cdots\}$, *if*

$$\|x_{k_i} + d_{k_i} - x^*\| = o(\|x_{k_i} - x^*\|), \quad k_i \to \infty, \tag{12.7.12}$$

then we have that

$$x_{k_i+1} = x_{k_i} + d_{k_i} \tag{12.7.13}$$

for all large i.

Proof. Without loss of generality, we assume that $k_i \geq k'$. For simplicity of notation, we substitute k_i by j. From the descriptions of the algorithm, we only need to show that

$$\Phi_{j,0}(1) - \Phi_{j,0}(0) - \mu\Phi'_{j,0}(0) < 0. \tag{12.7.14}$$

It follows from (12.7.10) that

$$\Phi_{j,0}(1) = f(x_j + d_j) - \lambda(x_j + d_j)^T c(x_j + d_j) + \frac{1}{2}\bar{\sigma}\|c(x_j + d_j)\|_2^2. \tag{12.7.15}$$

Because $f(x)$ is twice continuously differentiable, we have that

$$\begin{aligned} f(x_j + d_j) &= f(x_j) + \frac{1}{2}d_j^T[g_j + g(x_j + d_j)] + o(\|d_j\|_2^2) \\ &= f(x_j) + \frac{1}{2}d_j^T[g_j + g(x^*)] + o(\|d_j\|_2^2). \end{aligned} \tag{12.7.16}$$

Also, we can obtain similar formulae as (12.7.16) for $c_i(x_j + d_j)$. Substituting all these formulae into (12.7.15), we obtain that

$$\begin{aligned} \Phi_{j,0}(1) \quad - \quad & \Phi_{j,0}(0) = \frac{1}{2}d_j^T[g_j + g(x^*)] \\ & -\lambda(x_j + d_j)^T\left[c_j + \frac{1}{2}A_j^T d_j + \frac{1}{2}A(x^*)^T d_j\right] \\ & - \left[-\lambda_j^T c_j + \frac{1}{2}\bar{\sigma}\|c_j\|_2^2\right] + o(\|d_j\|_2^2) \\ = \quad & \frac{1}{2}\Phi'_{j,0}(0) + \frac{1}{2}d_j^T[g(x^*) - A(x^*)\lambda(x_j + d_j)] \\ & + o(\|d_j)\|_2^2) = \frac{1}{2}\Phi'_{j,0}(0) + o(\|d_j\|_2^2). \end{aligned} \tag{12.7.17}$$

It is not difficult to show there exists a positive constant $\bar{\eta}$ such that

$$\Phi'_{k,i}(0) \leq -\bar{\eta}\|d_k\|_2^2 \tag{12.7.18}$$

holds for all k and i. From (12.7.17), (12.7.18) and $\mu < \frac{1}{2}$, we can see that (12.7.14) holds for sufficiently large $j = k_i$. Thus, the lemma is true. □

A direct corollary of the above result is the superlinear convergence of the algorithm, which we write as follows.

Theorem 12.7.5 *Suppose that the assumptions of Lemma 12.7.2 are satisfied, and assume that the sequence $\{x_k\}$ generated by Algorithm 12.7.1 converges to x^*. If*

$$\lim_{k\to\infty} \frac{\|x_k + d_k - x^*\|}{\|x_k - x^*\|} = 0, \tag{12.7.19}$$

then we have that $x_{k+1} = x_k + d_k$ for all sufficiently large k, which implies that $\{x_k\}$ superlinearly converges to x^*.

12.8 Reduced Hessian Matrix Method

The reduced Hessian matrix method was also developed from the Lagrange-Newton method. A fundamental idea of the reduced Hessian matrix method is that only part of the Hessian matrix of the Lagrangian function is used so that the method requires less storage and computing costs in each iteration.

Consider the equality constrained problem (12.1.1) and (12.1.2). Denoting the Lagrange-Newton step by $(d_k, (\delta\lambda)_k)$, it follows from (12.1.6) that

$$\begin{bmatrix} W(x_k,\lambda_k) & -A(x_k) \\ -A(x_k)^T & 0 \end{bmatrix} \begin{bmatrix} d_k \\ (\delta\lambda)_k \end{bmatrix} = -\begin{bmatrix} \nabla f(x_k) - A(x_k)\lambda_k \\ -c(x_k) \end{bmatrix}. \tag{12.8.1}$$

Using the notations

$$\begin{aligned} W_k &= W(x_k,\lambda_k), && (12.8.2)\\ A_k &= A(x_k) = \nabla c(x_k)^T, && (12.8.3)\\ g_k &= \nabla f(x_k), && (12.8.4)\\ c_k &= c(x_k), && (12.8.5)\\ \hat{\lambda}_k &= \lambda_k + (\delta\lambda)_k, && (12.8.6) \end{aligned}$$

we can rewrite (12.8.1) as

$$\begin{bmatrix} W_k & -A_k \\ -A_k^T & 0 \end{bmatrix} \begin{bmatrix} d_k \\ \hat{\lambda}_k \end{bmatrix} = \begin{bmatrix} -g_k \\ c_k \end{bmatrix}. \tag{12.8.7}$$

Let the QR factorization of A_k be

$$A_k = [Y_k\ Z_k] \begin{bmatrix} R_k \\ 0 \end{bmatrix}, \tag{12.8.8}$$

then linear system (12.8.7) can be written in the following form:

$$\begin{bmatrix} Y_k^T W_k Y_k & Y_k^T W_k Z_k & -R_k \\ Z_k^T W_k Y_k & Z_k^T W_k Z_k & 0 \\ -R_k^T & 0 & 0 \end{bmatrix} \begin{bmatrix} p_k \\ q_k \\ \hat{\lambda}_k \end{bmatrix} = \begin{bmatrix} -Y_k^T g_k \\ -Z_k^T g_k \\ c_k \end{bmatrix}, \tag{12.8.9}$$

where

$$p_k = Y_k^T d_k, \tag{12.8.10}$$

$$q_k = Z_k^T d_k. \tag{12.8.11}$$

It is obvious that p_k and q_k are the projections of d_k to the range space of A_k^T and the null space of A_k^T. Because (12.8.9) has a block triangle form, we can easily solve p_k, q_k and $\hat{\lambda}_k$ in turns:

$$R_k^T p_k = -c_k, \tag{12.8.12}$$

$$(Z_k^T W_k Z_k) q_k = -Z_k^T g_k - Z_k^T W_k Y_k p_k, \tag{12.8.13}$$

$$R_k \hat{\lambda}_k = Y_k^T g_k + Y_k^T W_k (Y_k p_k + Z_k q_k). \tag{12.8.14}$$

If we consider only the last two lines in the linear system (12.8.9), we obtain a linear system independent of λ:

$$\begin{bmatrix} Z_k^T W_k Y_k & Z_k^T W_k Z_k \\ -R_k^T & 0 \end{bmatrix} \begin{bmatrix} p_k \\ q_k \end{bmatrix} = \begin{bmatrix} -Z_k^T g_k \\ c_k \end{bmatrix}, \tag{12.8.15}$$

which is essentially

$$\begin{bmatrix} Z_k^T W_k \\ -A_k^T \end{bmatrix} d_k = \begin{bmatrix} -Z_k^T g_k \\ c_k \end{bmatrix}. \tag{12.8.16}$$

Nocedal and Overton [232] suggests that the matrix $Z_k^T W_k$ be replaced by a quasi-Newton matrix B_k, namely at each iteration the line search direction d_k is obtained by solving the linear system

$$\begin{bmatrix} B_k \\ -A_k^T \end{bmatrix} d = \begin{bmatrix} -Z_k^T g_k \\ c_k \end{bmatrix}, \tag{12.8.17}$$

where $B_k \in \Re^{(n-m)\times n}$ is an approximation to $Z_k^T W_k$. We can apply Broyden's nonsymmetric rank-one formula to update B_k, that is

$$B_{k+1} = B_k + \frac{(y_k - B_k s_k) s_k^T}{s_k^T s_k}, \tag{12.8.18}$$

where

$$
\begin{aligned}
s_k &= x_{k+1} - x_k, && (12.8.19)\\
y_k &= Z_{k+1}^T g_{k+1} - Z_k^T g_k. && (12.8.20)
\end{aligned}
$$

Because $Z_k^T W_k$ is a one-side reduced Hessian matrix, this method is also called the one-side reduced Hessian matrix method. Under certain conditions, Nocedal and Overton [232] proved that the one-side reduced Hessian matrix method using (12.8.18)-(12.8.20) is locally superlinearly convergent.

If we use a symmetric matrix $B_k \in \Re^{(n-m)\times(n-m)}$ to substitute for $Z_k^T W_k Z_k$, and a zero matrix to replace $Z_k^T W_k Y_k$, we can see that (12.8.15) yields that

$$
\begin{bmatrix} 0 & B_k \\ -R_k^T & 0 \end{bmatrix} \begin{bmatrix} p_k \\ q_k \end{bmatrix} = \begin{bmatrix} -Z_k^T g_k \\ c_k \end{bmatrix}. \tag{12.8.21}
$$

One reason for doing so is a fact discovered by Powell [274] that the SQP method converges 2-step Q-superlinearly

$$
\lim_{k\to\infty} \frac{\|x_{k+1} - x^*\|}{\|x_{k-1} - x^*\|} = 0 \tag{12.8.22}
$$

provided $Y_k^T W_k Z_k$ is bounded. Another reason is that when all iteration points are feasible, we have $p_k = 0$, the value of $Z_k^T W_k Y_k$ does not alter q_k. For linearly constrained problems, all iteration points $x_k (k \geq k_0)$ are feasible if the initial point x_{k_0} is feasible. An advantage of updating $Z_k^T W_k Z_k$ instead of $Z_k^T W_k$ is that $Z_k^T W_k Z_k$ is a square matrix and it is symmetric positive definite near the solution where the second-order sufficient conditions hold. Therefore, we can use positive definite matrices to approximate it, such as the BFGS update:

$$
B_{k+1} = B_k - \frac{B_k s_k s_k^T B_k}{s_k^T B_k s_k} + \frac{y_k y_k^T}{s_k^T y_k}, \tag{12.8.23}
$$

where

$$
\begin{aligned}
s_k &= Z_k^T (x_{k+1} - x_k), && (12.8.24)\\
y_k &= Z_{k+1}^T g_{k+1} - Z_k^T g_k. && (12.8.25)
\end{aligned}
$$

We can write (12.8.21) in the equivalent form

$$
\begin{bmatrix} B_k Z_k^T \\ -A_k^T \end{bmatrix} d_k = \begin{bmatrix} -Z_k^T g_k \\ c_k \end{bmatrix}. \tag{12.8.26}
$$

Because the matrix B_k tries to approximate $Z_k^T W_k Z_k$, which is a two-side reduced Hessian matrix, the method using search direction d_k defined by (12.8.26) is called the two-side reduced Hessian matrix method. Such a method is two-step superlinearly convergent near the solution.

Theorem 12.8.1 *Let d_k be defined by (12.8.26). If $x_{k+1} = x_k + d_k$, $x_k \to x^*$, $A(x^*)$ is full column rank, second-order sufficient conditions hold at x^*, and $\|B_k^{-1}\|$ is bounded uniformly and satisfies*

$$\lim_{k\to\infty} \frac{\|[B_k - Z(x^*)^T W(x^*, \lambda^*) Z(x^*)] Z_k^T d_k\|}{\|d_k\|} = 0, \tag{12.8.27}$$

then the sequence converges 2-step Q-superlinearly:

$$\lim_{k\to\infty} \frac{\|x_{k+1} - x^*\|}{\|x_{k-1} - x^*\|} = 0. \tag{12.8.28}$$

Proof. It follows from (12.8.26) that

$$\begin{aligned} B_k Z_k^T d_k &= -Z_k^T g_k = -Z_k^T [g_k - A_k \lambda^*] \\ &= -Z_k^T W(x^*, \lambda^*)(x_k - x^*) + O(\|x_k - x^*\|^2). \end{aligned} \tag{12.8.29}$$

Thus, we have that

$$\begin{aligned} [B_k &- Z(x^*)^T W(x^*, \lambda^*) Z(x^*)] Z_k^T d_k \\ &= -Z_k^T W(x^*, \lambda^*)(x_k - x^*) - Z(x^*)^T W(x^*, \lambda^*) d_k \\ &\quad + O(\|x_k - x^*\|^2) + O(\|Y(x^*)^T d_k\|) + o(\|d_k\|) \\ &= -Z_k^T W(x^*, \lambda^*)(x_k + d_k - x^*) \\ &\qquad + O(\|x_k - x^*\|^2 + \|Y(x^*)^T d_k\|) + o(\|d_k\|). \end{aligned} \tag{12.8.30}$$

Therefore, based on the assumption (12.8.27), it follows from the above inequality that

$$Z_k^T W(x^*, \lambda^*)(x_k + d_k - x^*) = o(\|x_k - x^*\| + \|d_k\|) + O(\|Y(x^*)^T d_k\|). \tag{12.8.31}$$

The definition of d_k implies that

$$A_k^T (x_k + d_k - x^*) = O(\|x_k - x^*\|^2). \tag{12.8.32}$$

Because $A(x^*)$ is full column rank, we have that

$$\|Y(x^*)^T d_k\| = O(\|c(x_k)\|) = O(\|d_{k-1}\|^2). \tag{12.8.33}$$

Using (12.8.31) and (12.8.32), we see that

$$\begin{bmatrix} Z_k^T W(x^*, \lambda^*) \\ A_k^T \end{bmatrix} (x_k + d_k - x^*) = o(\|x_k - x^*\| + \|d_k\|) + o(\|d_{k-1}\|). \quad (12.8.34)$$

Observing the assumption that B_k^{-1} is uniformly bounded, and using (12.8.26), we can show that

$$\|d_k\| = O(\|x_k - x^*\|), \quad (12.8.35)$$

which indicates that $\|x_k - x^*\| \leq \|x_{k-1} - x^*\| + \|d_{k-1}\| = O(\|x_{k-1} - x^*\|)$. Thus, it follows from (12.8.34) that

$$\begin{bmatrix} Z_k^T W(x^*, \lambda^*) \\ A_k^T \end{bmatrix} (x_k + d_k - x^*) = o(\|x_{k-1} - x^*\|). \quad (12.8.36)$$

Similar to (12.3.19), we can prove that the matrix

$$\begin{bmatrix} Z(x^*)^T W(x^*, \lambda^*) \\ A(x^*)^T \end{bmatrix} \quad (12.8.37)$$

is nonsingular. Therefore, it follows from (12.8.36) that

$$\|x_k + d_k - x^*\| = o(\|x_{k-1} - x^*\|), \quad (12.8.38)$$

which shows that the theorem is true. $\Box$

The 2-step superlinearly convergence result of the two-side reduced Hessian matrix method can not be improved. In fact, an example given by Yuan [370] shows that it is possible to show that

$$\begin{aligned} \|x_{2k+1} - x^*\|_\infty &= \|x_{2k} - x^*\|_\infty, & (12.8.39) \\ \|x_{2k+2} - x^*\|_\infty &= \|x_{2k+1} - x^*\|_\infty^2, & (12.8.40) \end{aligned}$$

which reveals the "one-step fast one-step slow" behaviour of the two-side reduced Hessian matrix method, and it shows that it is impossible to establish a one-step Q-superlinearly convergence result. A similar example was also given by Byrd [40] independently.

Exercises

1. Use the Lagrange-Newton method to solve Rosenbrock's problem:

$$\begin{aligned} \min \quad & (1 - x_1)^2 \\ \text{s.t.} \quad & x_2 - x_1^2 = 0 \end{aligned}$$

with initial point $(0.8, 0.6)^T$ and $\lambda = 1.0$. Give the first three iterations.

2. The damped BFGS update (12.2.39) uses $\bar{y}_k$ which is a linear combination of y_k and $B_k s_k$. Consider the case generating $\bar{y}_k$ by linear combinations of y_k and s_k. What are the advantages and disadvantages of using s_k instead of $B_k s_k$?

3. Prove Corollary 12.3.4.

4. If the QP subproblem in a SQP method is infeasible, one way to overcome this difficulty is to consider the subproblem

$$\begin{array}{ll} \min_{d\in\Re^n,\theta\in[0,1]} & g_k^T d + \frac{1}{2} d^T B_k d + \sigma(1-\theta)^2 \\ \text{s.t.} & a_i(x_k)^T d + \theta c_i(x_k) = 0, \quad i \in E, \\ & a_i(x_k)^T d + \theta c_i(x_k) \geq 0, \quad i \in I. \end{array}$$

Let $(d(\sigma), \theta(\sigma)$ be the solution of the above QP subproblem. Prove that $\theta(\sigma)$ is non-decreasing as σ increases. Discuss the case when $\theta(\sigma) = 0$ for all $\sigma > 0$.

5. Consider application of the SQP method to the following problem:

$$\begin{array}{ll} \min & -x_1 + 10(x_1^2 + x_2^2) \\ \text{s.t.} & x_1^2 + x_2^2 = 1. \end{array}$$

Give the point $\bar{x} = (\cos(\theta), \sin(\theta))^T$ and calculate d by solving the QP subproblem with $B = I$. Assume θ is very small and show that

$$P_\sigma(\bar{x}) < P_\sigma(\bar{x} + d)$$

for any $\sigma \geq 0$, where $P_\sigma(x)$ is the L_1 exact penalty function. Calculate a second-order correction step $\hat{d}$ and verify that

$$P_\sigma(\bar{x}) > P_\sigma(\bar{x} + d + \hat{d}).$$

6. Prove that the Watchdog Technique (Algorithm 12.5.1) can overcome the Maratos Effect.

7. Apply the two-sided reduced Hessian method to the following problem:

$$\begin{aligned}
\min \quad & \frac{1}{2}x_2^2 - x_1x_2 + \frac{1}{6(1-x_2)^3}\Big[-4(x_2-x_1)^3 - 6(x_2-x_1)^2(x_1-x_2^2) \\
& -12(x_2-x_1)(x_1-x_2^2)^2 - 17(x_1-x_2^2)^3 + 3\frac{(x_1-x_2^2)^4}{1-x_2}\Big] \\
\text{s.t.} \quad & x_1 + \frac{1}{(1-x_2)^2}[(x_2-x_1)^2 + (x_2-x_1)(x_1-x_2^2) + 2(x_1-x_2^2)^2] = 0
\end{aligned}$$

with initial point (ϵ, ϵ), where $\epsilon > 0$ is a very small positive number. You will find the iterates converge to the solution $(0, 0)$ in the one-fast-one-slow pattern.